

PAULA FUDOLIG

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RESEARCH INTERESTS

Multi-wavelength studies of black hole – galaxy co-evolution.

EDUCATION

George Mason University

Doctor of Philosophy in Physics

Aug. 2023 – Present

Fairfax, VA

Boston University

Bachelor of Arts in Astronomy and Physics

Aug. 2019 – May 2023

Boston, MA

PROJECT EXPERIENCE

Radio Emission in Radio-Quiet Quasars

Graduate research, in collaboration with the National Radio Astronomy Observatory

May 2025 – Present

Fairfax, VA

- VLBA data reduction in **AIPS**, imaging, and analysis of optically-selected quasar observations at 6 GHz to trace the origin of observed radio-quiet emission.

Varstrometry for Dual AGN using Radio Interferometry

Graduate research, in collaboration with the US Naval Research Lab

June 2023 – Present

Fairfax, VA

- VLBA data reduction in **CASA**, imaging, and analysis of radio-loud quasar observations at 2 GHz and 8 GHz to investigate varstrometry+radio interferometry as a viable systematic detection method for multi-AGN.

Large VLBA BEAM-ME

Boston University Institute of Astrophysical Research

Jan. 2020 – May 2023

Boston, MA

- VLBA data reduction in **Difmap** of monthly blazar observations to monitor changing brightness and polarizations in blazar jets and their shockwaves and model jet activity over time.
- Determined and compared significant blazar features to investigate correlations between radio jet activity and gamma-ray events.

Patatrack Pixel Tracking

European Council for Nuclear Research (CERN)

Jan. 2022 – Aug. 2022

Geneva, Switzerland

- Tested Patatrack's 4 portability libraries of track reconstruction algorithms across 3 GPU devices on a varying amount of threads to evaluate performance, using CUDA in C++ and Python.

SOFIA 3-Micron Dust Emission Studies

NASA Goddard Space Flight Center

June 2020 – Aug. 2020

Greenbelt, MD

- Fitted 3-5 micron wavelength emission spectra in Python from polycyclic aromatic hydrocarbons surrounding a sample of planetary nebulae to study the relationship between hydrocarbon features and planetary nebulae ages.

PROPOSALS & GRANTS

Tracing the Origin of AGN Emission in Variable Radio-Quiet Quasars

Semester 2026A

- Principal investigator.

VLBA Characterization of Astrometrically-Variable Quasars

Semester 2024B

- Principal investigator.

Undergraduate Research Opportunities Program Grant

June 2022 – Aug. 2022

- > \$4000 in funding to extend internship at CERN.

PUBLICATIONS

Schwartzman, E., **Fudolig, P.**, et al. Varstrometry for Dual AGN using Radio Interferometry: VaDAR with the VLBA, 2025, ApJ, 987, 200

CONFERENCES, COLLOQUIA, & PRESENTATIONS

NRAO Socorro Lunch Talks <i>Talk title: A Tale of Two Variabilities: VaDAR and the Brightening J0822+4553</i>	Jan. 21, 2026 Socorro, NM
DC Astrophysics Graduate Student Conference <i>Talk title: Tracing the Origins of Radio Emission in Radio-Quiet Quasars</i>	Aug. 14-15, 2025 Washington, D.C.
245th American Astronomical Society Meeting <i>Talk title: A VLBA View of VaDAR</i>	Jan. 12-16, 2025 National Harbor, MD
11th Annual Mid-Atlantic Radio-Loud AGN Meeting <i>Talk title: A VLBA View of VaDAR</i>	November 8, 2024 Baltimore, MD
DC Astrophysics Graduate Student Conference <i>Talk title: A VLBA View of VaDAR</i>	Aug. 15-16, 2024 Washington, D.C.
10th Annual Mid-Atlantic Radio-Loud AGN Meeting <i>Talk title: VaDAR: Varstrometry for Dual AGN using Radio Interferometry</i>	October 12, 2023 Fairfax, VA
Boston University Undergraduate Research Symposium <i>Poster title: Pixel-Tracking in CERN's Compact Muon Solenoid</i>	Oct. 21, 2022 Boston, MA
NASA Goddard Summer Poster Session <i>Talk title: Comparing Fits of PAH Emission</i>	August 4, 2020 Greenbelt, MD

SERVICE

SPECTRUM Leader <i>Organization dedicated to empowerment and community building in STEM students</i>	March 2024 – Present Fairfax, VA
• Created and implemented a series of lunch workshops promoting the professional development of graduate and undergraduate students within the GMU College of Science. • Created and implemented a mentorship program for incoming graduate students, including developing mentor/mentee applications, matching mentoring pairs, advertising the opportunity, and organizing related social events. Resulted in a 100% retention rate upon its first implementation in Spring 2024.	
Student Advisor <i>GMU College of Science Dean's Student Advisory Board</i>	August 2025 – Present Fairfax, VA
• Shared insights and feedback on the student experience, representing graduate student needs and interests. • Collaborated with the Dean's office on college-level initiatives and improvements. • Fostered a stronger sense of community across programs and disciplines within the College of Science.	
Physics Department Representative <i>GMU's Graduate and Professional Student Association (GAPSA)</i>	August 2024 – May 2025 Fairfax, VA
• Advocated for fellow Physics graduate students on topics including graduate tuition and fees, fellowship opportunities, student life, and campus events. • Collaborated with other graduate student representatives from all departments on University-level initiatives. • Communicated important information on departmental changes and solicited feedback for upcoming votes.	
Peer Mentor <i>BU's PeeRs for Incoming Student Mentorship (PRISM)</i>	Sep. 2021 – May 2023 Boston, MA
• Guided first-year students by providing departmental and campus resources, creating four-year plans, sharing personal experiences, and providing support throughout the academic year.	
Founder of BU Chapter of Society of Women in Space Exploration <i>Organization supporting the inclusion of women and minorities interested in space-related fields</i>	Aug. 2020 – May 2023 Boston, MA
• As Founder: authored organization's constitution and bylaws, assembled an executive board, recruited a faculty advisor, and advertised on campus to build membership. • As President: led biweekly general member meetings, involving devising group activities, planning on-campus events, and networking in the Boston area for collaborations. Also led executive board meetings, delegating tasks and overseeing management of responsibilities.	

RELATED EXPERIENCE

Graduate Teaching Assistant

Aug. 2023 – Present

- Led undergraduate labs in Physics I/II and Astronomy I at George Mason University.

Astronomy on Tap Speaker

November 18, 2025

- Presented "Hunting for Dual AGN with VaDAR" to the general public.

Observation Time on Perkins Telescope

March 8-10, 2023

- Three nights operating Boston University's Perkins Telescope at Anderson Mesa Observatory in Flagstaff, AZ.

SCIENTIFIC & TECHNICAL SKILLS

Programming: Python, C++, CUDA, ROOT, Git, LaTeX, Ubuntu, Jupyter Notebook, Mathematica, Bash/Shell

Astronomical: NRAO CASA, NRAO AIPS, Caltech Difmap, PyBDSF, TopCat, SAOImage DS9, Maxim DL

World Languages: English (native), Spanish (advanced), French (intermediate), Tagalog (basic)